1.

$\hat{SE} = \sqrt{\frac{S^2}{n}} = \frac{S}{\sqrt{n}} = \frac{10}{10} = 1$

$H_0: \mu = 50$
$H_1: \mu \neq 50$

$Z = \frac{52-50}{1} = 2, \quad P = 2\Phi(-2) = 0.046$

$\alpha = .1 \rightarrow 0.046 \leq 0.1$, so we reject the null hypothesis

$\alpha = .05 \rightarrow 0.046 \leq 0.05$, so we reject the null hypothesis

$\alpha = .01 \rightarrow 0.046 > 0.01$, so we fail to reject the null hypothesis

95% CI $\rightarrow$ $C = 1.96$, $\bar{X} = 52$, $\hat{SE} = 1$

$52 - 1.96(1) = 50.04 \quad 52 + 1.96 = 53.96$

$[50.04, 53.96] \rightarrow$ 95% CI

2.

$\hat{SE} = \frac{S}{\sqrt{n}} = \frac{3}{\sqrt{36}} = \frac{3}{6} = \frac{1}{2}$

95% CI $\rightarrow$ $C = 1.96$, $\bar{X} = 11.5$, $\hat{SE} = \frac{1}{2}$

$11.5 - 1.96\left(\frac{1}{2}\right) = 10.52 \quad 11.5 + 1.96\left(\frac{1}{2}\right) = 12.48$

$[10.52, 12.98] \rightarrow$ 95% CI

$Z = \frac{11.5-12}{\frac{1}{2}} = -1, \quad P = 2\Phi(-1) = 0.317$

Since 12 is contained in our 45% confidence interval, it is a plausible value for our population mean. This means we do not have evidence to reject the null hypothesis that $\mu \neq 12$. Equivalently, the p-value is 0.317 and since $0.317 > 0.05$, we fail to reject the null hypothesis that $\mu \neq 12$. In context, we do not have enough to believe that the mean lifespan differs from the brochure.

3.

$\widehat{SE}_A = \frac{S}{\sqrt{n}} = \frac{5}{\sqrt{10}} = \frac{4}{5}$ $\qquad \widehat{SE}_{A-B} = \sqrt{\left(\frac{4}{5}\right)^2 + \left(\frac{3}{10}\right)^2} = \sqrt{1.13}$

$\widehat{SE}_B = \frac{S}{\sqrt{n}} = \frac{3}{10} = \frac{3}{10}$

$H_0 : \mu_A = \mu_B$ $\qquad Z = \frac{(\bar{X}_A - \bar{X}_B) - 0}{\widehat{SE}_{A-B}} = \frac{(25-22)-0}{\sqrt{1.13}} \approx 2.82$

$H_1 : \mu_A \neq \mu_B$ $\qquad P = 2\,\Phi(-2.82) \approx 0.005$

95% CI: $c = 1.96$, $\bar{X}_A - \bar{X}_B = 3$, $\widehat{SE}_{A-B} = \sqrt{1.13}$

$3 - 1.96\left(\sqrt{1.13}\right) \approx 0.92$ $\qquad 3 + 1.96\left(\sqrt{1.13}\right) \approx 5.08$

$\therefore [0.92, 5.08]$

The 95% CI does not include 0, thus 0 is not a plausible value for A and B. Thus, we reject the null hypothesis $\mu_A = \mu_B$. Equivalently, the P-value is 0.005, which is well below the threshold of $\alpha = 0.05$. Thus, we should reject the null hypothesis; there is evidence to believe that the average prices of homes in neighborhood A differ from the average prices of homes in neighborhood B.